

COMP 6721 - Project proposal - Group A

A. Problem Statement and Application

Nowadays, people are paying more and more attention to the security of their personal information by installing antivirus software. However, malware continuously evolves as attackers develop new techniques to bypass detection systems [1], which makes traditional methods such as file extension block-list and dynamic file analysis struggle to cope with those evolves. In such a situation, deep learning should be implemented to assist with antivirus software. Deep learning particularly convolutional neural networks (CNN), offers a significant advantage over traditional methods due to automatic learning. Unlike traditional rule-based methods, which rely on predefined rules and behaviors, CNN can be trained by large datasets and quickly adapt to new malware. This makes deep learning highly suitable for detecting evolving threats, as it can continuously improve using new data. This project aims to develop a reliable system that uses CNN, learns from the malware dataset, and is able to classify and detect various malware. By training on portable executable (PE) files in byteplot image format, the system should be able to use visual patterns to offer a more intuitive and efficient detection method than the traditional method.

B. Image Dataset Selection

In this project, we are using Blended Malware Image Dataset, Malware Images, and Malware-Byteplot Images. All three datasets are from Kaggle. We will use the Blended Malware Image Dataset[2] as the large dataset. This dataset consists of 31 classes, 9868 images in total. All images are formatted in PNG format with 224*224 pixel resolution. Most of the class has 350 images each, while the following six classes have varying numbers.

- Alueron.gen!J: 173 images, C2LOP.gen!J: 175 images, Dialplatform.B: 152 images, Fakerean: 306 images, Lolyda.AA1: 153 images, Lolyda.AA2 :159 images

Malware Images[3] has 17 classes with 125004 images. All images are 224*224 pixel PNG files. As this dataset is highly skewed to a few specific classes, we will preprocess this dataset to fit with the requirement. This dataset will be used as our moderate dataset. Lastly, Malware-Byteplot Images[4] dataset consists of only two classes which are Benign and Malicious. It has 60337 images in total. All images are 224*224 pixel PNG files. It is also slightly skewed towards the Malicious class. Hence, we will also adjust it to fit the requirements. This dataset will be used as our small dataset.

C. Possible Methodology

This methodology outlines the use of CNNs: for our malware detection system, focusing on 3 pre-trained models: ResNet [4], EfficientNet [5], and DenseNet [6]. ResNet utilizes residual connections, which help in learning deeper networks and mitigating the vanishing gradient problem. We will use ResNet-18, a simpler variation of the architecture, to strike a balance between depth and efficiency. EfficientNet-B0 employs compound scaling, optimizing width, depth, and resolution, allowing us to process large datasets efficiently. DenseNet, known for its densely connected layers, strengthens feature propagation and reuse, which enhances detection accuracy. DenseNet-121 will be used for its simplicity and efficiency.

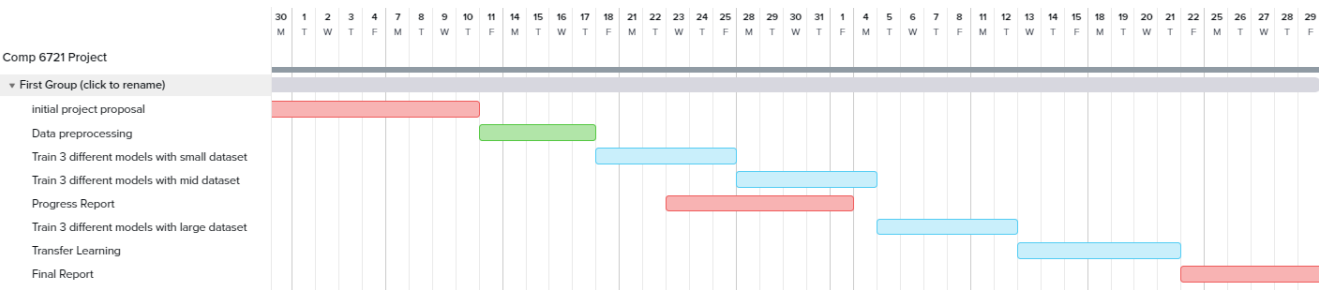
In the **Model Training** phase, we will train 9 models: 3 different CNN architectures across 3 datasets (large, moderate, and small). Each model will utilize transfer learning by initializing with pre-trained ImageNet weights. 2 models will undergo transfer learning, with pre-trained weights adjusted to fit our specific malware detection task, while others will train from scratch. We will apply techniques like data augmentation (e.g., rotation, flipping, brightness adjustments) to increase dataset diversity, fine-tune hyperparameters using validation data, and implement learning rate scheduling (e.g., step decay) to optimize learning. Regularization methods such as dropout and L2 regularization will be employed to minimize overfitting.

To **Evaluate the models**, we will use accuracy to measure overall performance, precision to assess the proportion of true positives, and recall to determine how well the models identify actual malware. F1-score will provide a balanced view by combining precision and recall. A confusion matrix will give detailed insights into performance across all classes. We will also use AUC-ROC to evaluate class discrimination and monitor false positive and false negative rates (FPR and FNR) to ensure the system minimizes misclassifications. These metrics will guide us in developing an efficient and accurate malware detection system.

By utilizing these architectures, training methods, and evaluation metrics, we aim to develop a robust and accurate malware detection system capable of identifying various types of malware from image representations of executable files.

D. Gantt Chart

The following image is the Gantt chart for this project. On the left, each deliverable in this project has been divided into tasks. The right-hand side of this image shows the estimated start time and deadline for each task.



References

[1] Molinari S, Packer J. *MALWARE SCIENCE a Comprehensive Guide to Detection, Analysis, and Compliance*. Packt Publishing Ltd.; 2023. Accessed October 7, 2024. <https://www.oreilly.com/library/view/-/9781804618646/>

[2] Gaurav Pendharkar. 2022 *Blended Malware Image Dataset*. Kaggle. Available online: <https://www.kaggle.com/datasets/gauravpendharkar/blended-malware-image-dataset?resource=download>

[3] Walt. 2022 *Malware Images*. Kaggle. Available online: <https://www.kaggle.com/datasets/walt30/malware-images/>

[4] Ajit Varma. 2024 *Malware-Byteplot Images*. Kaggle. Available online: <https://www.kaggle.com/datasets/ajit22/malware-byteplot-images>

[5] He, K., Zhang, X., Ren, S., & Sun, J. 2016. Deep residual learning for image recognition. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 770-778).

[6] Tan, M., & Le, Q. 2019. EfficientNet: Rethinking model scaling for convolutional neural networks. In International Conference on Machine Learning (pp. 6105-6114). PMLR.

[7] Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 4700-4708).